

TrES-2b

R-2536

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_100925 · created 2026-08-25T23:36:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455464.709698 +/- 3.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.021
Transit depth (Rp/Rs)^2	1.62 +/- 0.55 [%]
Orbital Inclination (inc)	84.34 +/- 1.03
Airmass coefficient 1 (a1)	1.032 +/- 0.013
Airmass coefficient 2 (a2)	-0.021 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	#2 - [140, 338]
Optimal Aperture	3.63
Optimal Annulus	10.87
Transit Duration (day)	0.079 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.127 $\pm$ 0.021 vs 0.1254 (deviation 0.08 σ)
Duration fit vs published	0.079 d vs 0.0746 d (deviation 0.26 σ)
Mid-time O–C	0.1 min
Depth SNR	6.0
Residual scatter	1.31 %

Light curve

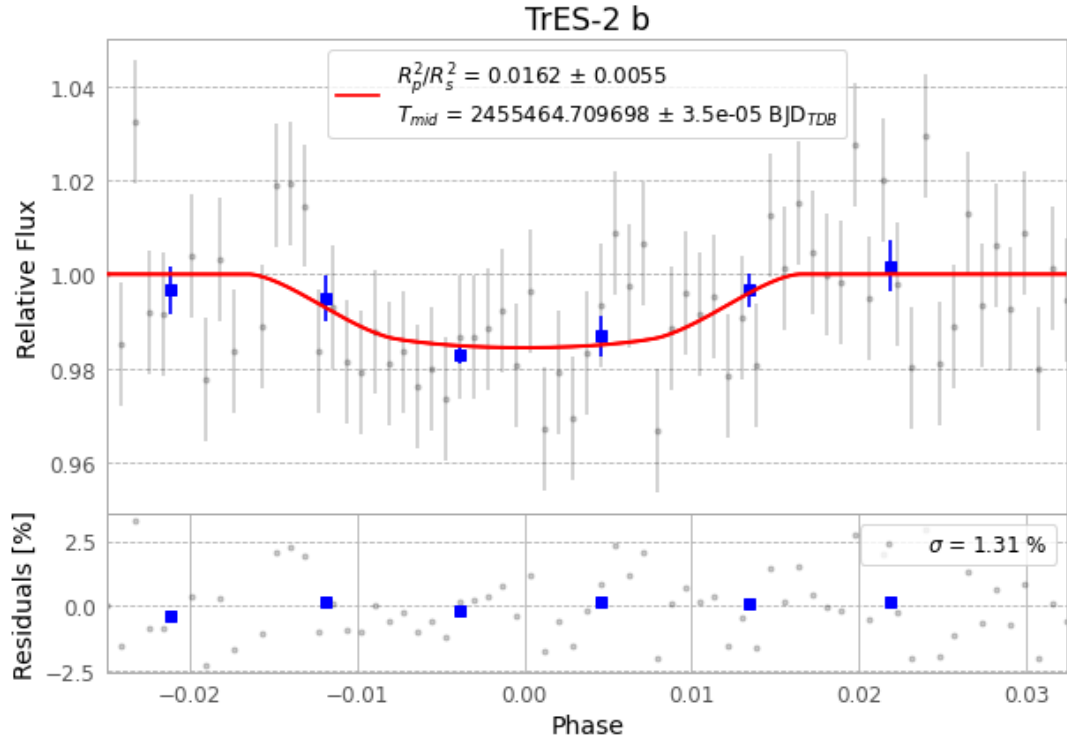


Figure 1 — FinalLightCurve_TrES-2 b_2010-09-24.png (SHA-256 25d62d217b935a98...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e8a165955a2fc6a4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

69 FITS frames (45 MB), TRES-2100925033014.FITS → TRES-2100925065414.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-100925.log (74f3676720f8...), FinalLightCurve_TrES-2 b_2010-09-24.png (25d62d217b93...), FinalLightCurve_TrES-2 b_2010-09-24.csv (98f3a5cd980f...)

Compute resources

Wall time 97.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:36Z.